CS 228: Logic in CS Autumn 2025

Mid Semester Examination

Total Marks: 80 marks *17 September 2025*

Read the following instructions carefully.

- The examination is of three hours' duration and is for a total of 80 marks.
- The question paper has a total of 6 pages, counting this page. Verify that your copy contains all pages.
- Write your ROLL NUMBER on your answer sheet.
- Unless asked for explicitly, you may cite results/proofs covered in class without reproducing them.
- When asked to prove something, give a complete formal proof with suitable explanation. Similarly, if asked to disprove give a counter-example and argue why it is in-fact a counter-example. Informal and incomplete arguments will lose marks.
- If you need to make any assumptions, state them clearly. Marks will be given only if your assumption is valid.
- Answers written with pencil won't be evaluated. No cribs will be handled over this later on.
- **DO NOT COPY SOLUTIONS FROM OTHERS**. Any instance of Academic dishonesty will be taken seriously and reported to DADAC.
- Throughout this question paper wff denotes a well formed formula, in respective logics.
- We assume that we are always given with the equality predicates when dealing with First Order Logic, unless the question explicitly says so.
- When drawing an automata, make sure to clearly indicate the start state. Automata without start state will be considered invalid and will be awarded *zero* marks. No cribs will be handled over this.
- Make sure to draw the automata neatly and show the transitions clearly. You are bound to lose marks in case of ambiguity. Show the accepting states precisely.
- Distribute your time judiciously across the questions. Some questions may require more time; allocate your efforts accordingly.
- Instructions specific to individual questions are provided alongside them. These must be followed strictly. Any cribs or grievance arising from non-compliance will not be entertained later.

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1. **[10 marks]**(**Recalling Basics of Propositional Logic and FOL**)

State TRUE or FALSE for each of the below. **1 mark for each correct answer**. Give a one or two sentence explanation about your answer. **Answers without justification will fetch *zero* marks, even if your answer is correct.**

Make sure that you write TRUE *or* FALSE *legibly. In case of any ambiguous answers, your answer will be treated as incorrect.*

(a) Consider the signature $\tau = (P^2, f^1, g^2, c, d, e)$ consisting of a binary relation P, a unary function f, a binary function g as well as constants c, d, e. Recall the definition of well formed formulae (wff) defined in class.

i. $P(x, c) \wedge f(c)$ is a wff

ii. $f(g(c, f(d))) = g(x, y)$ is a wff

(b) The formulae $\exists x \forall y (R(x, y) \wedge P(x) \vee x < y)$ and $\forall y (R(x, y) \wedge P(x) \vee x < y)$ are equivalent.

(c) The formula $\forall x \exists y [P(x, y) \to \forall y Q(y)]$ is a rectified formula.

(d) Every sentence in First Order Logic has an equivalent formula in Skolem Normal Form.

(e) The satisfiability problem for First Order Logic is semi-decidable in general, however, there exist fragments of FO where the satisfiability problem is decidable.

(f) Given a Horn formula of length n, Horn SAT takes at least n^2 steps to conclude.

(g) Recall how we defined wff in FOL in class. According to what we have defined, any formula in FOL is written on a non-empty signature.

(h) Let F be a formula in propositional logic written in CNF. Then satisfiability checking for F can be done in polynomial time; however, there is no known polynomial time procedure for checking its validity.

(i) The language $(ab)^*$ is FO-definable while the language $(aa)^*$ is not.

Solution

The rubrics is given along with the solutions:

(a) i. False. f is a function, thus conjunction of a function with predicate does not form a wff.

ii. True. The left side and right side of the equality are valid terms respecting the arities.

(b) False. Suitable counterexamples have been awarded marks. For example, empty universe. No marks if you have given assignment for y (or x for the first formula) since they are bound variables.

(c) False. y has to be renamed.

(d) False. Skolem Form is equisatisfiable not equivalent.

(e) True. 0.5 marks for describing how ground resolution results in semi-decidability terminates for UNSAT but not for SAT). 0.5 marks for example of decidability of FO over words using the FO to Automata translation we did in class. No marks for examples like UNSAT formulas or only mentioning propositional logic/CNF form without showing how these can be fragments of FOL i.e. Quantifier-less logic.

(f) False. It takes at most n^2 steps to conclude. Any suitable counter examples are awarded marks. Ex, it can finish in n steps, if there no unit clause.

(g) False. Any valid examples, like $x = y$ is a valid wff over an empty signature.

(h) False. Validity is in P but SAT is NP for CNF. 0.5 marks for mentioning each complexity.

(i) True. $(ab)^*$ can be characterized as words starting with a, ending with b where every a is followed by a b and every b should be followed by an a if it exists. ($(ab)^*$ also includes ε). We have seen in class that $(aa)^*$ is not FO definable. 0.5 marks for correct characterization of $(ab)^*$ and 0.5 for mentioning even string of a's is not FO-definable.

2. **[20 marks]** (**Propositional Logic : Good Old Completeness**) Let Vars be a fixed set of variables. Denote by $\mathcal{P}$, the set of all propositional logic formulas, constructed using symbols in the set $\{\bot, \top, \neg, \wedge, \vee, \rightarrow\}$ in addition to variables from Vars and parantheses. Denote by $\mathcal{Q}$ the set of all propositional logic formulas, constructed using symbols in the set $\{\bot, \rightarrow\}$ in addition to variables from Vars and parantheses.

For two sets of formulae $\mathcal{A}$ and $\mathcal{B}$, we say $\mathcal{A} \sqsubseteq \mathcal{B}$ if for all formulae $\varphi_A \in \mathcal{A}$, there exists a formula $\varphi_B \in \mathcal{B}$ such that φ_A and φ_B are *semantically equivalent*. (Recall that two formulae are *semantically equivalent* if they are over the same set of variables and has the same truth values for all assignments over these variables.) If $\mathcal{A} \sqsubseteq \mathcal{B}$ and $\mathcal{B} \sqsubseteq \mathcal{A}$, we say $\mathcal{A} \equiv \mathcal{B}$.

(a) **[5 marks]** Show that $\mathcal{P} \equiv \mathcal{Q}$. Since $\mathcal{Q}$ is a subset of $\mathcal{P}$, it is obvious that $\mathcal{Q} \sqsubseteq \mathcal{P}$. So, it suffices for you to show that $\mathcal{P} \sqsubseteq \mathcal{Q}$.

(b) **[5 marks]** Your solution to the previous subquestion should convince you that any propositional logic formula can be converted to a semantically equivalent one using only $\rightarrow$ and $\bot$. We now extend the notion of natural deduction proofs to derive sequents using formulae in $\mathcal{Q}$. For this we use the $\rightarrow_i$, $\rightarrow_e$ and $\bot_e$ rules of natural deduction you have seen in class. Additionally you are provided with a special rule called $(\rightarrow \bot)_e$ defined as below:

$$(\rightarrow \bot)_e : \frac{(\phi \rightarrow \bot) \rightarrow \psi \quad \phi \rightarrow \zeta \quad \psi \rightarrow \zeta}{\zeta}$$

Using **ONLY** the above four rules ($\rightarrow_i$, $\rightarrow_e$, $\bot_e$ and $(\rightarrow \bot)_e$), prove the following sequent:

$$(\phi \rightarrow \bot) \rightarrow \psi, \phi \rightarrow \zeta \vdash (\psi \rightarrow \bot) \rightarrow \zeta$$

If you use any other rule other than the above, you will get *zero* marks.

(c) **[10 marks]** The above four rules i.e, $\rightarrow_i$, $\rightarrow_e$, $\bot_e$ and $(\rightarrow \bot)_e$ form a proof system to derive formulas in $\mathcal{Q}$. Do you think this proof system is complete with respect to $\mathcal{Q}$? In other words, given two formula $\phi, \psi \in \mathcal{Q}$ such that $\phi \models \psi$, is it always possible to prove the sequent $\phi \vdash \psi$ using only the above four rules? Argue formally.

Solution

(a) We prove this by structural induction. Base case is trivial.

For this we use the following semantic equivalences:

$$\begin{aligned}
\top &\equiv p \to p \\
\neg\varphi &\equiv (\varphi \to \bot) \\
\varphi_1 \vee \varphi_2 &\equiv (\varphi_1 \to \bot) \to \varphi_2 \\
\varphi_1 \wedge \varphi_2 &\equiv (\varphi_1 \to (\varphi_2 \to \bot)) \to \bot
\end{aligned}$$

Then by structural induction, it is clear that for any $\varphi \in \mathcal{P}$, we have some formula in $\mathcal{Q}$ preserving semantic equivalence.

PS: This is a very concise version exhibiting just the semantic equivalences. You are expected to give complete structural induction proof for full merit.

(b) Given below is the required proof:

1.	$(\phi \to \bot) \to \psi$	premise
2.	$\phi \to \zeta$	premise
3.	$\psi \to \bot$	assumption
4.	ψ	assumption
5.	$\bot$	$\to_e$ 3,4
6.	ζ	$\bot_e$ 5
7.	$\psi \to \zeta$	$\to_i$ 4-6
8.	ζ	$(\to \bot)_e$ 1,2,7
9.	$(\psi \to \bot) \to \zeta$	$\to_i$ 3-8

(c) We need to show that if $\phi \models \psi$, then $\phi \vdash \psi$, i.e, we can contsruct a proof for ψ starting with ϕ as the premise. Now observe that via semantics of $\to$, $\phi \models \psi$ is same as $\models (\phi \to \psi)$

So, we will first prove that if $\models \varphi$ then we can construct a proof for $\vdash \varphi$ using the new proof system. This is an adaption of the completeness proof done in class. We begin by proving the following Lemma:

Lemma 1 *Let φ be a formula over variables $p_1, \ldots, p_n$ and let $\hat{p}_i$ denote p_i if p_i is true in line, ℓ of the truth table and $\hat{p}_i$ denote $p_i \to \bot$ if p_i is false in line ℓ. Then,*

- $\hat{p_1}, \ldots, \hat{p_n} \vdash \varphi$ *if* φ *evaluates to true in line* ℓ
- $\hat{p_1}, \ldots, \hat{p_n} \vdash (\varphi \to \bot)$ *if* φ *evaluates to false in line* ℓ

The proof is by structural induction. The base case is when $\varphi = p$ or $\varphi = \bot$ which can be proven easily (as $p \vdash p$ and $\vdash (\bot \to \bot)$. Since $\varphi \in \mathcal{Q}$, the only connective is $\to$. Thus for structural induction we only need to consider the case, $\varphi = \varphi_1 \to \varphi_2$.

Assume that the lemma holds true for φ_1 and φ_2. We will now prove that the lemma holds true for φ.

Case 1. Consider any line, ℓ of the truth table where φ evaluates to false. This can happen only when φ_1 evaluates to true and φ_2 evaluates to false in that line. By induction hypothesis, $\hat{p_1}, \ldots, \hat{p_n} \vdash \varphi_1$ and $\hat{p_1}, \ldots, \hat{p_n} \vdash (\varphi_2 \to \bot)$. Then we have:

1.	$\hat{p_1}, \ldots, \hat{p_n}$	premise
2.	$\varphi_1 \to \varphi_2$	assumption
3.	φ_1	as $\hat{p_1}, \ldots, \hat{p_n} \vdash \varphi_1$
4.	φ_2	$\to_e$ 2,3
5.	$\varphi_2 \to \bot$	as $\hat{p_1}, \ldots, \hat{p_n} \vdash (\varphi_2 \to \bot)$
6.	$\bot$	$\to_e$ 4,5
7.	$(\varphi_1 \to \varphi_2) \to \bot$	$\to_i$ 2-6

Case 2. Consider any line, ℓ of the truth table where φ_2 evaluates to true. Then, φ evaluates to true in that line. By induction hypothesis, $\hat{p_1}, \ldots, \hat{p_n} \vdash \varphi_2$. Then we have:

1.	$\hat{p_1}, \ldots, \hat{p_n}$	premise
2.	φ_1	assumption
3.	φ_2	as $\hat{p_1}, \ldots, \hat{p_n} \vdash \varphi_2$
4.	$\varphi_1 \to \varphi_2$	$\to_i$ 2-3

Case 3. Consider any line, ℓ of the truth table where φ_1 and φ_2 both evaluates to false. Then, φ evaluates to true in that line. By induction hypothesis, $\hat{p_1}, \ldots, \hat{p_n} \vdash (\varphi_2 \to \bot)$ and $\hat{p_1}, \ldots, \hat{p_n} \vdash (\varphi_1 \to \bot)$. Then we have:

1.	$\hat{p_1}, \ldots, \hat{p_n}$	premise
2.	φ_1	assumption
3.	$\varphi_1 \to \bot$	as $\hat{p_1}, \ldots, \hat{p_n} \vdash (\varphi_1 \to \bot)$
4.	$\bot$	$\to_e$ 2,3
5.	φ_2	$\bot_e$ 4
6.	$\varphi_1 \to \varphi_2$	$\to_i$ 2-5

To complete the proof we would need another lemma which is proved below:

Lemma 2 *Let $p, \varphi \in \mathcal{Q}$ and $\Gamma \subset \mathcal{Q}$ be a set of premises, then under the new proof system if $p \cup \Gamma \vdash \varphi$ and $(p \to \bot) \cup \Gamma \vdash \varphi$, then $\Gamma \vdash \varphi$.*

This can be seen via the following proof in our new proof system:

1.	Γ	premise
2.	p	assumption
3.	φ	as $p \cup \Gamma \vdash \varphi$
4.	$p \to \varphi$	$\to_i$ 2-3
5.	$p \to \bot$	assumption
6.	φ	as $(p \to \bot) \cup \Gamma \vdash \varphi$
7.	$(p \to \bot) \to \varphi$	$\to_i$ 5-6
8.	φ	assumption
9.	φ	
10.	$\varphi \to \varphi$	$\to_i$ 8-9
11.	φ	$(\to \bot)_e$ 4,7,10

Consider the truth table of φ where $\models \varphi$. From Lemma 1, we have 2^n proofs of the form, $\hat{p_1}, \ldots, \hat{p_n} \vdash \varphi$ corresponding to each row of the truth table. ($\models \varphi$, each row of the truth table will be true.) Now we will combine these proofs to obtain a single proof for $\vdash \varphi$. Observe that we have 2^{n-1} proofs for $\hat{p}_1, \ldots, \hat{p}_{n-1}, p_n \vdash \varphi$ and another 2^{n-1} proofs for $\hat{p}_1, \ldots, \hat{p}_{n-1}, (p_n \to \bot) \vdash \varphi$. Each of these 2^{n-1} proofs correspond to different possibilities of $\hat{p}_1, \ldots, \hat{p}_{n-1}$.

For each of these possibilities,

$$\hat{p_1}, \ldots, \hat{p}_{n-1}, p_n \vdash \varphi \text{ and } \hat{p_1}, \ldots, \hat{p}_{n-1}, (p_n \to \bot) \vdash \varphi \implies \hat{p_1}, \ldots, \hat{p}_{n-1} \vdash \varphi$$

due to Lemma 2. Thus we get 2^{n-1} proofs of the form $\hat{p_1}, \ldots, \hat{p}_{n-1} \vdash \varphi$. Inductively, keep removing each of the $\hat{p}_i$ by repeatedly applying Lemma 2 to construct the final proof for $\vdash \varphi$.

Now, returning back to our original question, if $\varphi := (\phi \to \psi)$, we have $\models \varphi$ and we have derived the proof for $\vdash \varphi$, i.e, $\vdash (\phi \to \psi)$. Use this proof to obtain the proof for $\phi \vdash \psi$ by pushing the ϕ in the assumption to the premise.

Rubrics

(a)
- Solutions with proper, correct and complete structural induction gets 5 marks.
- Many people has incorrectly done structural induction, but I have awarded 4 marks if all the necessary equivalences are clearly mentioned.
- For each wrong equivalence, 1 mark has been deducted. I have given marks even if the equivalence is expressed in terms of $\neg$ and $\neg$ is derived correctly. If $\neg$ is wrong, everything depending on it is awarded 0 mark.
- We have been quite liberal grading this part this time. DO NOT TAKE THIS FOR GRANTED. We won't be compromising on formality and precison from next time.
- If you got 4 marks and didn't do structural induction precisely, DO NOT crib.

(b)
- 5 marks for correct and complete proof.
- If there is some mistake in between the proof, 0 marks. Cribs will be handled only if you have used all the rules correctly and have not used any semantic equivalences.
- 0 marks if your proof contains any rule other than is specified or any formula not in $\mathcal{Q}$.
- Partial marks awarded only when minor mistakes which does not affect the proof structure is made.

(c)
- 10 marks for complete proof exhibiting all the steps in model solution.
- Partial Marks are awarded as per merit. Talk to the TA during cribs in case of any doubt.
- Some people has tried to derive the rules of Natural deduction to prove the completeness, but all of them could not explain why that is sufficient and how it is directly related to completeness. Most of these just derived some common semantic equivalences and could not argue why they in fact forms the rules of Natural Deduction. Any such proof is awarded at most 5 marks (depending on correctness, relevance and completeness).

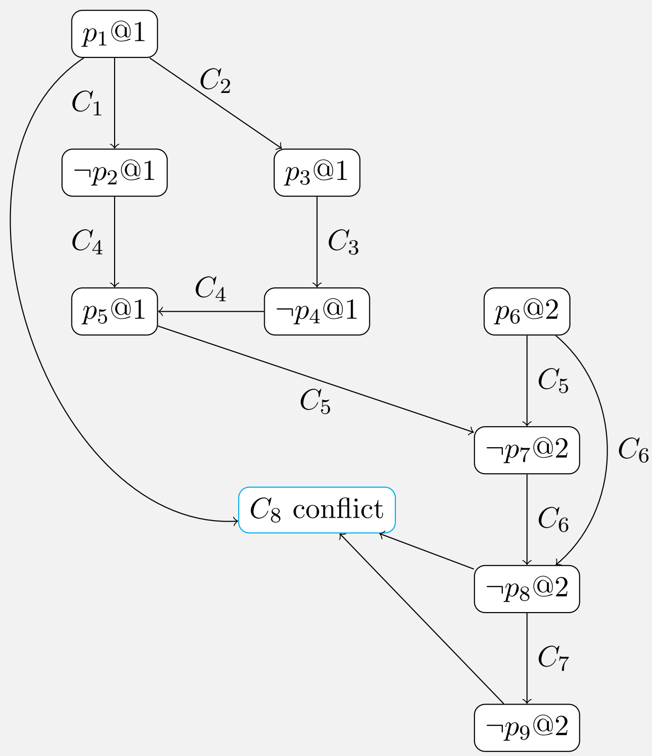
3. **[6 marks]** (**Propositional Logic : Satisfiability Checking using DPLL**) Simulate a run of the DPLL algorithm with clause learning (as done in class and the slides) on the formula with the following clauses. Whenever you need to make a decision, use the decision strategy *assign true to the currently unassigned variable of least index.* (For example if p_2, p_4 and p_5 are unassigned, in the decision step, you should choose p_2 and assign it true.)

   Draw the implication graph as you solve, clearly showing your decision and propagation nodes.

$$
\begin{aligned}
C_1 &= (\neg p_1 \vee \neg p_2)\\
C_2 &= (\neg p_1 \vee p_3)\\
C_3 &= (\neg p_3 \vee \neg p_4)\\
C_4 &= (p_2 \vee p_4 \vee p_5)\\
C_5 &= (\neg p_5 \vee \neg p_6 \vee \neg p_7)\\
C_6 &= (\neg p_6 \vee p_7 \vee \neg p_8)\\
C_7 &= (p_8 \vee \neg p_9)\\
C_8 &= (p_8 \vee p_9 \vee \neg p_1)
\end{aligned}
$$

Solution

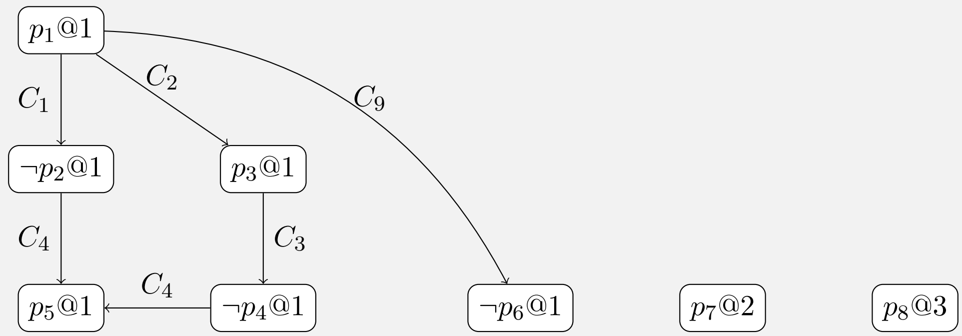
Run 1



Since clause learning adds the disjunction of the negations of all decision literals involved in the conflict, the learned clause is

$$\boxed{C_9 = \neg p_1 \vee \neg p_6}$$

Run 2 with the learnt clause



The assignment generated for the propositional variables $p_1, \dots, p_9$ is:

$$(p_1, p_2, \dots, p_9) = (\top, \bot, \top, \bot, \top, \bot, \top, \top, x), \quad x \in \{\top, \bot\}$$

where the i-th entry corresponds to the value assigned to p_i, for $i = 1, \dots, 9$.

Rubrics

(a) Marks allocation:

- 2 marks — First run of DPLL
- 1 mark — The learnt clause
- 3 marks — Second DPLL run

(b) Deductions and guidelines:

- 0 marks awarded for a DPLL run if the specified decision policy is not used.
- If one part of the question is incorrect (wrong decision clause learnt or incorrect DPLL run), subsequent parts are not corrected.
- 0.5 marks deducted for minor mistakes, such as missing edges or incorrect notation of decision levels.
- Maximum 3 marks awarded if a linear graph is drawn instead of the correct implication graph.

4. **[8 marks] (FOL with and without equality)**

(a) [**6 marks**] For this sub-part, consider **FOL without equality**. Let F be a satisfiable FO-sentence. Let $\mathcal{A}$ be a structure satisfying F with $|U^{\mathcal{A}}| = n \geq 1$. Prove formally, that for every $m \geq n$, there exists a structure $\mathcal{B}_m$ satisfying F with $|U^{\mathcal{B}_m}| = m$.

(b) [**2 marks**] For this sub-part, consider **FOL with equality**. For every $n \geq 1$, find a satisfiable FO-sentence F_n such that for every structure $\mathcal{A}$, $\mathcal{A}$ satisfies F_n if and only if $0 < |U^{\mathcal{A}}| \leq n$. Justify your choice of F_n.

Solution

(a) Construction: Choose an element $a \in U^{\mathcal{A}}$ (exists, since $|U^{\mathcal{A}}| \geq 1$) and construct a new structure $\mathcal{B}$ with $|U^{\mathcal{B}}| = |U^{\mathcal{A}}| + 1$ as follows.

- $U^{\mathcal{B}} = U^{\mathcal{A}} \cup \{b\}$ where $b \notin U^{\mathcal{A}}$ is the new element
- Define a onto mapping $f : U^{\mathcal{B}} \to U^{\mathcal{A}}$ such that $f(x) = x$ if $x \in U^{\mathcal{A}}$ and $f(b) = a$. Extend this mapping naturally to sequences of elements as $f : (U^{\mathcal{B}})^k \to (U^{\mathcal{A}})^k$ as $f(x_1, \ldots, x_k) = (f(x_1), \ldots, f(x_k))$. This will be used in extending the relations, functions, and assignments.
- For all predicates and functions in the signature (implicit from the formula F), extend them to make b mimic the behaviour of a, i.e.,
  * For all relations R of k-arity in the signature, $R^{\mathcal{B}} = \{x \in (U^{\mathcal{B}})^k | f(x) \in R^{\mathcal{A}}\}$
  * For all functions f of k-arity in the signature, $f^{\mathcal{B}}$ is extended as: $f^{\mathcal{B}}(x) = f^{\mathcal{A}}(f(x))$ for all $x \in (U^{\mathcal{B}})^k$

Prove by structural induction on the formula φ that:
For all assignments α to the free variables in φ,
$\mathcal{A} \models_\alpha \varphi \iff (\mathcal{B} \models_\beta \varphi$ for all β such that some a's in α are replaced by b's in $\beta)$

- Atomic formulae
- $\wedge, \vee, \neg, \rightarrow$
- $\forall, \exists$

Hence, for any FO-sentence F, $\mathcal{A} \models F \iff \mathcal{B} \models F$.

(b) $\varphi_1 = (\exists x(x = x))$, $\varphi_2 = \forall x_1 \cdots \forall x_{n+1} \left(\bigvee_{1 \leq i < j \leq n+1} (x_i = x_j)\right)$
$\varphi = \varphi_1 \wedge \varphi_2$
Justification: $\mathcal{A} \models \varphi_1 \iff |U^{\mathcal{A}}| > 0$ and $\mathcal{A} \models \varphi_2 \iff |U^{\mathcal{A}}| \leq n$ (where the contrapositive of the forward direction can be proved by assigning $(n+1)$ different elements to the variables x_1 to x_{n+1}, and the backward direction follows from Pigeonhole Principle).

Rubrics

(a)
- 2 marks - Construction
- 1.5 mark - Stating what we have to prove using structural induction
- 1 mark - Base case: Atomic formulae
- 0.5 mark - $\wedge, \vee, \neg, \rightarrow$
- 1 mark - $\forall, \exists$

The proof is not evaluated if the construction is not correct.

(b) 1 mark deducted if $|U^{\mathcal{A}}| > 0$ is not enforced or if justification is not written. Only 0.5 marks given - if only $|U| \leq n$ is enforced without justification.

5. **[13 marks]** (**The Decidability Quest : FOL to Automata**) Consider the signature $\tau = (<, Q_a, Q_b)$ for words, and the sentence

$$\varphi = \forall x(Q_a(x) \to \neg\exists y([x < y \wedge \neg\exists z(x < z \wedge z < y)] \wedge Q_b(y))$$

(a) **[1 mark]** What is $L(\varphi)$?

(b) **[12 marks]** Follow the steps done in class for FO–DFA conversion and construct the DFA A such that $L(\varphi) = L(A)$.
(***Hint:*** *If you wish, you can simplify φ to a formula ψ over the signature $\tau' = (S, Q_a, Q_b)$ such that $L(\psi) = L(\varphi)$, and then draw the equivalent DFA for ψ following all steps. In this case, you must write your simplified sentence ψ and justify (informally) why $L(\psi) = L(\varphi)$ before constructing the DFA.*)

You **should** clearly follow all the steps in the conversion as done in class and exhibit the intermediate automata.

Solution

(a) $L(\phi) = b^*a^*$

If the language was described, but regex wasn't mentioned $\to$ 0.5 marks

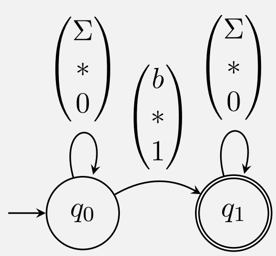
Correct regex mentioned $\to$ 1 marks

(b) $F(x, y) = (x < y \wedge \neg\exists z(x < z \wedge z < y)) \equiv S(x, y)$

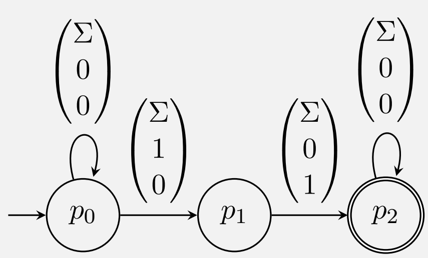
So, define ψ as

$$\psi = \forall x\,(Q_a(x) \to \neg\exists y[S(x, y) \wedge Q_b(y)])$$
$$\equiv \neg\exists x\,(Q_a(x) \wedge \exists y[S(x, y) \wedge Q_b(y)])$$

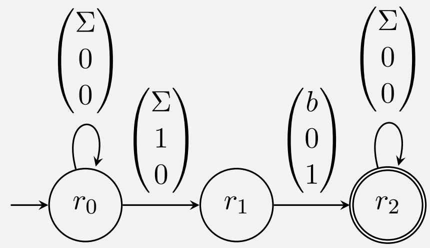
– $Q_b(y)$



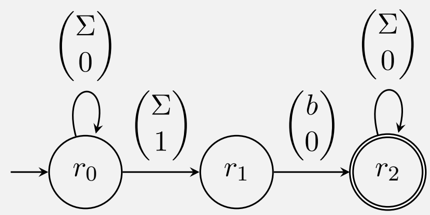
– $S(x, y)$



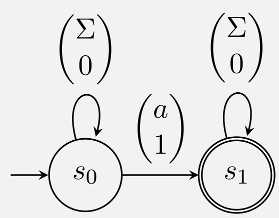
- $Q_b(y) \wedge S(x,y)$



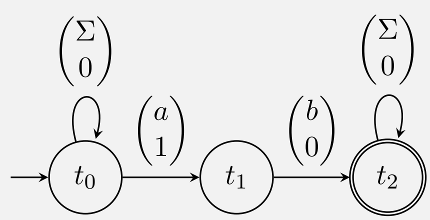
- $\exists y[S(x,y) \wedge Q_b(y)]$



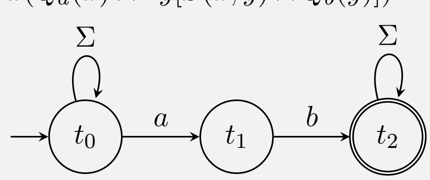
- $Q_a(x)$



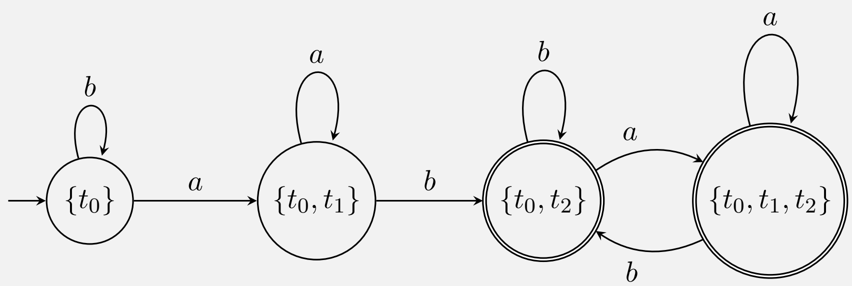
- $Q_a(x) \wedge \exists y[S(x,y) \wedge Q_b(y)]$



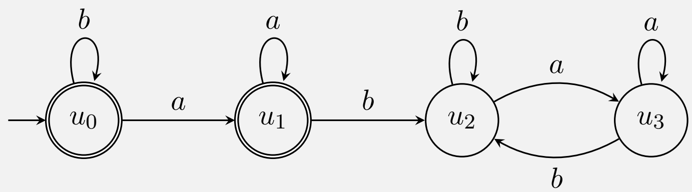
- $\exists x(Q_a(x) \wedge \exists y[S(x,y) \wedge Q_b(y)])$



- Determinizing



– $\neg\exists x(Q_a(x) \wedge \exists y[S(x,y) \wedge Q_b(y)])$

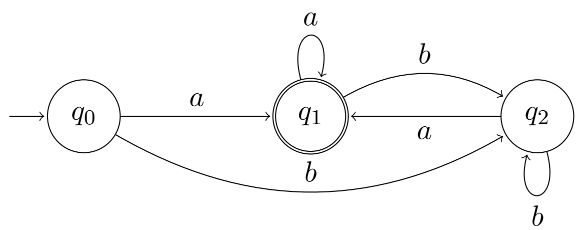


Note : You have received full marks only if all the steps are shown, and no state/ transition is missing. If you have made a mistake in some step, later steps may not be checked. Failure to indicate start/ final states carries a large penalty, and any cribs to consider automata without these will not be entertained later.

Drawing automata directly from the language accepted will fetch *zero* marks

6. **[15 marks]** (**DFAs : The Mirror of Noitaziminim**)

Malay and Yuvraj, two curious wizards at Hogwarts, sneak one night into the office of Prof. Siva, the renowned professor of Transcendental Automagica. In Prof. Siva's office, they meet a strange automaton A that appears in front of them (see below).



The automaton blinked and asked softly, "What brings you here at this hour?"

Yuvraj leaned closer and whispered, "This Saturday we face a trial in Automagica, and we fear Prof. Siva will make it quite tricky for us. Our friend Jigyasa told us about this office, saying we might find some hidden charms here to guide us."

The automaton sighed, "Ah, I understand. I do know where many secrets lie, but they are locked away inside a narrow vault. The trouble is, I am too heavy, burdened with extra states, and cannot reach them. If you could help me shed this weight, I might just be able to slip inside and help you."

Malay suggests: "We could try Abhilasha's *reducto* procedure, which will sever away your redundant states!" But the automaton trembles: "No, no! I'm too scared...it will hurt. There must be a gentler magic."

Seeing the *Mirror of Noitaziminim* in Siva's office, Yuvraj exclaims : "This can help you! Let's try the *double mirror-convergence* charm in front of the mirror and your essence will shine through in a leaner form."

Thus begins their magical quest, with each step a challenge. Please assist Malay, Yuvraj and the automaton by answering the following questions that Malay and Yuvraj pose:

(a) **[1 mark] Malay :** "Dear automaton A, before we begin, tell us, what is your language $L = L(A)$ that you accept? Explain to us informally, so we know who you truly are."

(b) **[1 mark] Yuvraj :** "And tell us this: Are you carrying a *phantom pair* of states? That is, do you have a pair of states q, q' that behave identically on all possible words w? Formally, are there q, q' such that for all $w \in \Sigma^*$, $\hat{\delta}(q, w) \in F$ iff $\hat{\delta}(q', w) \in F$? If yes, list all such possible pairs. If not, why do you say so?"

(c) **[2 marks] Malay :** "Please position yourself in front of the mirror so that the mirror charm works on you. The charm will make you accept the reverse of all words you once accepted. In effect, no states will be added to/removed from you. However, the mirror will rearrange your transitions and also change the start and accepting states suitably. Your new form will be called B, which accepts the reverse of all the words you initially accepted. Reveal your new form to us." Please draw B.

(d) **[1 mark] Yuvraj :** "Move out now. If you feel you have become non-deterministic after the first mirror charm, I will cast the *convergence charm* to make you deterministic and call your new deterministic form C. If you think your form B is deterministic, then the convergence charm will leave you unchanged." Draw the new form C.

(e) **[2 marks] Malay :** "Let us now apply the mirror charm a second time. Position yourself again in front of the mirror. Just as we got the form B from A, we will now obtain from C a new form which accepts the reverse of all words initially accepted in C. From C, this mirrored form will be called D." Draw D.

(f) **[2 marks] Yuvraj :** "Move out from the mirror now. If your form D is non-deterministic, I will cast the *convergence charm* again to help you regain determinism. Call the new form E. If the form D is already deterministic, then E is the same as D. Show us your form E." Draw E.

Automaton : I am tired, I have gone through enough charms. Can we please stop?

(g) **[1 mark] Malay :** "Okie, we are done. The power of the double mirror-convergence should have worked by now. How many states do you now have, E? Count carefully."

(h) **[1 mark] Yuvraj :** "And tell us, Is your language $L(E)$ the same as your original language $L(A)$? Explain to us informally, so we see whether your identity is restored."

(i) **[1 mark] Malay inspects closely:** "Are you carrying any phantom pairs, E? If yes, list them. If not, justify why no such pair exists anymore."

(j) **[3 marks] Finally, Yuvraj concludes:** "If your new form E has no phantom pair of states, but your original form A did, then did you figure how those vanished? What is this mysterious charm really doing to you?"

Automaton : Thanks guys, I feel really trim!

The automaton thinks of what just happened. It realizes that standing twice before the *Mirror of Noitaziminim*, and passing through the convergence fire each time, has purified it. Whatever fat it had is gone, yet its true self, the language it speaks remains intact.

With a lighter frame and a brighter glow, the automaton whispers, "You came here seeking charms for your trial in Automagica. In helping me, you have already found one. Remember this — *It is our choices that determine what we truly are, far more than our abilities.*"

Malay and Yuvraj smile at each other. They sneak out of Siva's office, hearts lighter than before, carrying no secret parchment, but instead a deeper magic : the knowledge that even the heaviest automaton can be made lean while keeping its soul.

Solution

(a) A accepts exactly those non-empty words over $\{a, b\}$ whose last symbol is a.

$$L(A) = \{\, w \in \{a, b\}^* \mid w \text{ ends with } a \,\} = \{a + b\}^* a.$$

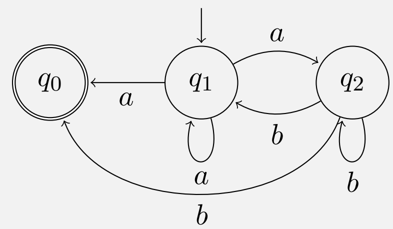
(The empty word ε is *not* accepted because the start state q_0 is non-accepting.)

(b) Yes. q_0 and q_2 are indistinguishable (a phantom pair).

Reason: after reading the first symbol both q_0 and q_2 behave the same (on a they both go to q_1, on b they both go to q_2), and for the empty word both are non-accepting. So the pair (q_0, q_2) is the only phantom pair. q_1 is accepting and hence distinguishable from the others.

(c) Reverse every transition of A; make the original accepting states the start states, and make the original start state the single accepting state.

B is:



(d) Start subset is $\{q_1\}$. Using the reversed transition relation:

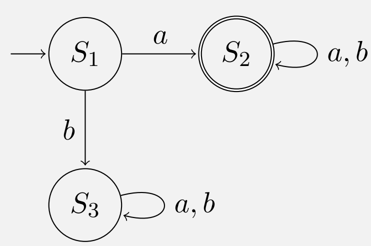
$$\begin{aligned} \delta_B(\{q_1\}, a) &= \{q_0, q_1, q_2\}, & \delta_B(\{q_1\}, b) &= \varnothing, \\ \delta_B(\{q_0, q_1, q_2\}, a) &= \{q_0, q_1, q_2\}, & \delta_B(\{q_0, q_1, q_2\}, b) &= \{q_0, q_1, q_2\}, \\ \delta_B(\varnothing, a) &= \varnothing, & \delta_B(\varnothing, b) &= \varnothing. \end{aligned}$$

Accepting subsets are those containing q_0 (since q_0 was the accepting state of B).

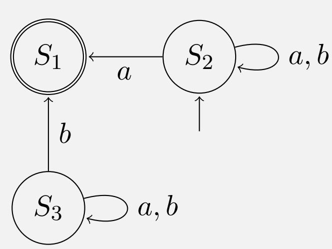
So the reachable DFA states are:

$$\begin{aligned} &S_1 = \{q_1\} \quad \text{(start, non-accepting)}, \\ &S_2 = \{q_0, q_1, q_2\} \quad \text{(accepting)}, \\ &S_3 = \varnothing \quad \text{(sink, non-accepting)}. \end{aligned}$$

We draw C using these names:



(e) Reverse every arrow of C and swap start/accept to get NFA D with start S_2 and unique accepting state S_1.
D is:



(f) Start subset is $\{S_2\}$. The transitions are:

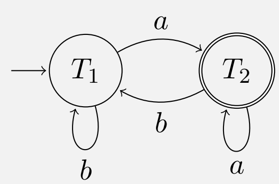
$$\begin{aligned} \delta_D(\{S_2\}, a) &= \{S_1, S_2\}, & \delta_D(\{S_2\}, b) &= \{S_2\}, \\ \delta_D(\{S_1, S_2\}, a) &= \{S_1, S_2\}, & \delta_D(\{S_1, S_2\}, b) &= \{S_2\}. \end{aligned}$$

(The subset $\varnothing$ is not reachable here.)

Accepting subsets are those containing S_1. The reachable subsets are:

$$T_1 = \{S_2\} \quad \text{(start, non-accepting)}, \qquad T_2 = \{S_1, S_2\} \quad \text{(accepting)}.$$

E is:



(g) E has $\boxed{2}$ states.

(h) Yes. The sequence of steps we performed (reverse, determinize, reverse, determinize) preserves the accepted language, as $\text{rev}(\text{rev}(L)) = L$.
Alternately, both A and E accept exactly the non-empty words ending in a. Hence $L(E) = L(A)$.

(i) No. E has two states: one is accepting (T_2) and the other is non-accepting (T_1). They are distinguishable by the empty word: T_2 accepts ε while T_1 does not — so they are not equivalent. Therefore E has no phantom pairs.

(j) The idea is this: suppose in the original DFA we have two different states, but from both of them you can follow exactly the same paths to reach an accepting state. These states are “phantom” duplicates — they behave differently in appearance but actually do the same job in the automaton.

Now, when we reverse the automaton, the accepting states become starting points. So if two states had the same set of paths to a final state originally, then after reversal they give rise to the same set of paths starting from the new initial state.

When we then determinize (do the subset construction), these two states naturally get merged into a single state, because the subset construction collects together all states that represent the same set of possibilities.

Repeating the process (reverse + determinize again) ensures that no such duplicates are left, and we end up with the smallest DFA possible.

So the “phantom pair” disappears not by magic, but simply because reversing + determinization forces states with identical future behaviour to be clubbed together automatically, yielding a minimal DFA.

Rubrics

Main rule: For parts (c), (d), (e), (f), if a student scores **0 in any subpart, the following subparts will not be checked.**

(a) 1 if correct; 0 otherwise.

(b) 1 if correct; 0 otherwise.

(c) 2 if correct; 1 for minor mistakes (e.g., making q_2 final); 0 otherwise.

(d) 1 if correct; 0.5 if missing transitions on trap state; 0 otherwise.

(e) 2 if correct automaton; 1 if state 3 is made final or no transitions on state 3; 0 if trap state is pruned or otherwise incorrect.

(f) 2 if correct automaton; 1 for minor mistakes; 0 otherwise.

(g) 1 if correct; 0 otherwise.

(h) 0.5 if "yes"+ 0.5 for correct justification.

(i) 0.5 if "no"+ 0.5 for correct justification.

(j) 3 for correct justification; 1–2 for partial justification.

7. **[6 marks]** (**Lest you complain there is no DFA problem**) Let L be a regular language over the alphabet $\{0,1\}$. Assuming an order $0 < 1$ between the symbols, define

$$L_i = \{w \mid w \in L, |w| = i\}, \quad \text{and} \quad w^i_{\text{min}} = \text{the lexicographically smallest word in } L_i$$

For example, if $L_3 = \{001, 010, 011\}$, then $w^3_{\text{min}} = 001$. Likewise if $L_4 = \{0110, 1010, 1100, 1101, 1111\}$, then $w^4_{\text{min}} = 0110$.

Define $L_{\text{min}} = \{w^i_{\text{min}} \mid i \in \mathbb{N}\}$. Show that L_{min} is regular.

Solution

Let DFA $A = \{Q_A, \Sigma = \{0,1\}, q_{0,A}, F_A, \delta_A\}$ be the automata s.t. $L = L(A)$.

Consider DFA B s.t. $L(B) = \{(\omega_1, \omega_2) \mid \omega_1, \omega_2 \in L,\ |\omega_1| = |\omega_2|\}$.

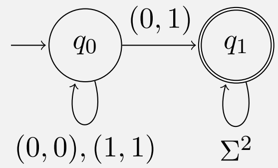
$$B = \{Q_A \times Q_A,\ \Sigma \times \Sigma,\ (q_{0,A}, q_{0,A}),\ F_A \times F_A,\ \delta_B\}$$

such that

$$\delta_B((q_1, q_2), (a, b)) = (\delta_A(q_1, a),\ \delta_A(q_2, b)).$$

Consider NFA C s.t. $L(C) = \{(\omega_1, \omega_2) \mid \omega_1, \omega_2 \in \Sigma^*,\ |\omega_1| = |\omega_2|,\ \omega_1 <_{lex} \omega_2\}$

$C:$



By the cross-product construction, we construct NFA $D = (Q_D, \Sigma^2, q_{0,D}, F_D, \delta_D)$ s.t. $L(D) = L(B) \cap L(C)$

So, $L(D) = \{(\omega_1, \omega_2) \mid \omega_1, \omega_2 \in L,\ |\omega_1| = |\omega_2|,\ \omega_1 <_{lex} \omega_2\}$

Finally, we project ω_1 out from D to get NFA $E = (Q_D, \Sigma, q_{0,D}, F_D, \delta_E)$ as follows:

$$\delta_E(q, a) = \delta_D(q, (a, 0)) \cup \delta_D(q, (a, 1)).$$

Now, language of E is s.t. $L(E) = \{\omega \mid \exists v \in L,\ v <_{lex} \omega,\ |v| = |\omega|\}$.

We construct NFA F by cross-product construction such that $L(F) = L(A) - L(E)$. Since the language of F is $\mathrm{L}_{\mathrm{min}}$, we can conclude that L_{min} is regular.

8. **[2 marks]** (**One Last Face**) Consider the alphabet $\Sigma = \{$☺, ☹$\}$. Write down a word (of length 7) over this alphabet representing the sequence of your emotions while solving questions 1-7.

Rubrics

Everyone who has attempted this question is given 2 marks. If you haven't received 2 marks despite attempting it, talk to the TAs during cribs.